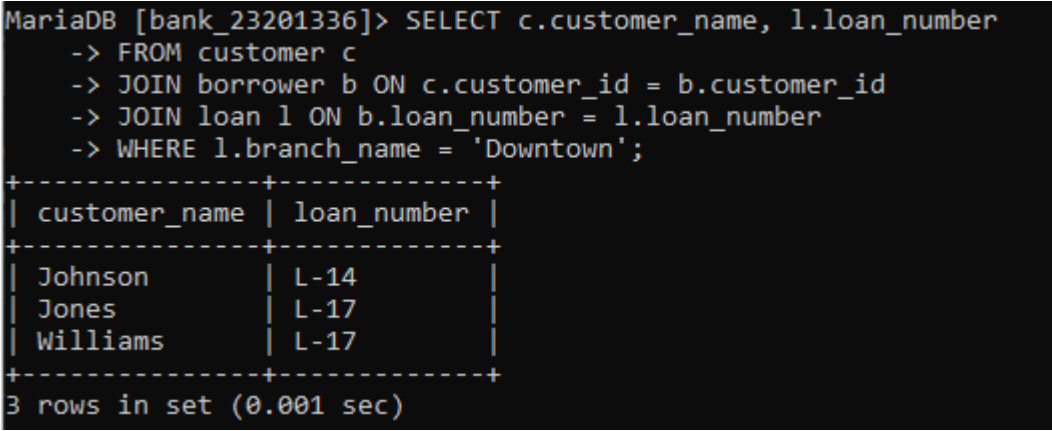
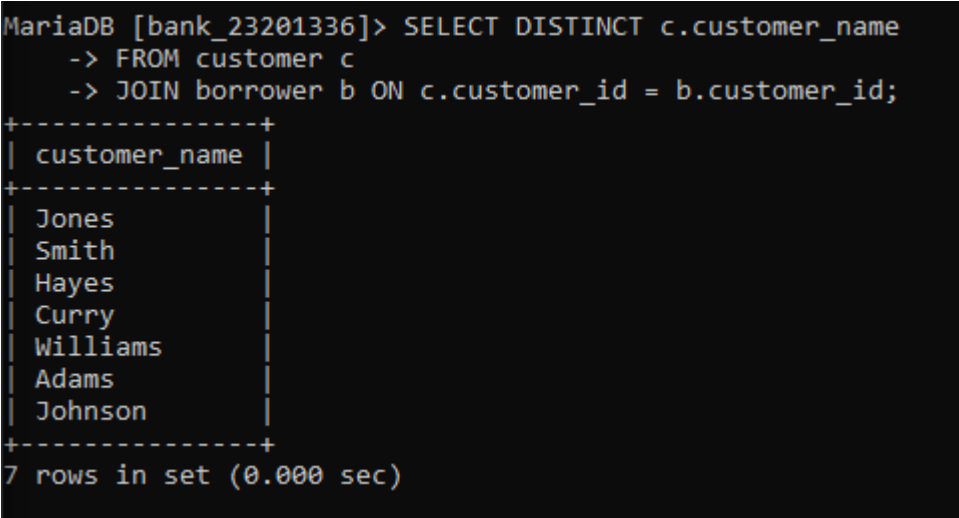


CSE370 : Database Systems

Assignment 03 | Spring 2026

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No 1 Query (as Plain Text)	SELECT c.customer_name, l.loan_number FROM customer c JOIN borrower b ON c.customer_id = b.customer_id JOIN loan l ON b.loan_number = l.loan_number WHERE l.branch_name = 'Downtown';
No 1 SS (of Query & Output in Shell)	 <pre>MariaDB [bank_23201336]> SELECT c.customer_name, l.loan_number -> FROM customer c -> JOIN borrower b ON c.customer_id = b.customer_id -> JOIN loan l ON b.loan_number = l.loan_number -> WHERE l.branch_name = 'Downtown'; +-----+-----+ customer_name loan_number +-----+-----+ Johnson L-14 Jones L-17 Williams L-17 +-----+-----+ 3 rows in set (0.001 sec)</pre>
No 2 Query (as Plain Text)	SELECT DISTINCT c.customer_name FROM customer c JOIN borrower b ON c.customer_id = b.customer_id;

No 2 SS (of Query & Output in Shell)	 <pre> MariaDB [bank_23201336]> SELECT DISTINCT c.customer_name -> FROM customer c -> JOIN borrower b ON c.customer_id = b.customer_id; +-----+ customer_name +-----+ Jones Smith Hayes Curry Williams Adams Johnson +-----+ 7 rows in set (0.000 sec) </pre>
No 3 Query (as Plain Text)	<pre> SELECT c.customer_name FROM customer c LEFT JOIN borrower b ON c.customer_id = b.customer_id WHERE b.loan_number IS NULL; </pre>

No 3 SS (of Query & Output in Shell)	<pre> MariaDB [bank_23201336]> SELECT c.customer_name -> FROM customer c -> LEFT JOIN borrower b ON c.customer_id = b.customer_id -> WHERE b.loan_number IS NULL;_ +-----+ customer_name +-----+ Lindsay Turner Glenn Brooks Green +-----+ 5 rows in set (0.000 sec) </pre>
No 4 Query (as Plain Text)	<pre> SELECT c1.customer_name AS Customer1, c2.customer_name AS Customer2, c1.customer_city AS City FROM customer c1 JOIN customer c2 ON c1.customer_city = c2.customer_city AND c1.customer_id < c2.customer_id; </pre>

No 4 SS
(of Query & Output
in Shell)

```
MariaDB [bank_23201336]> SELECT c1.customer_name AS Customer1, c2.customer_name AS Customer2, c1.customer_city AS City
-> FROM customer c1
-> JOIN customer c2
-> ON c1.customer_city = c2.customer_city
-> AND c1.customer_id < c2.customer_id;
+-----+-----+-----+
| Customer1 | Customer2 | City |
+-----+-----+-----+
| Jones     | Hayes     | Harrison |
| Smith     | Curry     | Rye      |
| Lindsay   | Adams     | Pittsfield |
| Turner    | Green     | Stamford |
+-----+-----+-----+
4 rows in set (0.000 sec)
```

No 5 Query
(as Plain Text)

```
SELECT branch_name,
ROUND(SUM(balance * 0.04), 2) AS Total_Interest
FROM account
GROUP BY branch_name;
```

No 5 SS
(of Query & Output
in Shell)

```
MariaDB [bank_23201336]> SELECT branch_name,  
-> ROUND(SUM(balance * 0.04), 2) AS Total_Interest  
-> FROM account  
-> GROUP BY branch_name;  
+-----+-----+  
| branch_name | Total_Interest |  
+-----+-----+  
| Brighton   |          66.00 |  
| Downtown   |          20.00 |  
| Mianus      |          28.00 |  
| Perryridge |          16.00 |  
| Redwood    |          28.00 |  
| Round Hill |          14.00 |  
+-----+-----+  
6 rows in set (0.000 sec)
```

No 6 Query
(as Plain Text)

```
SELECT c.customer_city, a.account_number  
FROM customer c  
JOIN depositor d ON c.customer_id = d.customer_id  
JOIN account a ON d.account_number = a.account_number  
WHERE a.balance = (  
SELECT MAX(a2.balance)  
FROM customer c2  
JOIN depositor d2 ON c2.customer_id = d2.customer_id  
JOIN account a2 ON d2.account_number = a2.account_number  
WHERE c2.customer_city = c.customer_city  
);
```

No 6 SS
(of Query & Output
in Shell)

```
MariaDB [bank_23201336]> SELECT c.customer_city, a.account_number
-> FROM customer c
-> JOIN depositor d ON c.customer_id = d.customer_id
-> JOIN account a ON d.account_number = a.account_number
-> WHERE a.balance = (
-> SELECT MAX(a2.balance)
-> FROM customer c2
-> JOIN depositor d2 ON c2.customer_id = d2.customer_id
-> JOIN account a2 ON d2.account_number = a2.account_number
-> WHERE c2.customer_city = c.customer_city
-> );
```

customer_city	account_number
Harrison	A-217
Rye	A-215
Pittsfield	A-222
Stamford	A-305
Palo Alto	A-201

```
5 rows in set (0.001 sec)
```

No 7 Query
(as Plain Text)

```
SELECT l.loan_number, l.amount, c.customer_name
FROM loan l
JOIN borrower b ON l.loan_number = b.loan_number
JOIN customer c ON b.customer_id = c.customer_id
ORDER BY l.amount DESC
LIMIT 5;
```

No 7 SS
(of Query & Output
in Shell)

```
MariaDB [bank_23201336]> SELECT l.loan_number, l.amount, c.customer_name
-> FROM loan l
-> JOIN borrower b ON l.loan_number = b.loan_number
-> JOIN customer c ON b.customer_id = c.customer_id
-> ORDER BY l.amount DESC
-> LIMIT 5;
```

loan_number	amount	customer_name
L-23	2000	Smith
L-14	1500	Johnson
L-15	1500	Hayes
L-16	1300	Adams
L-17	1000	Jones

5 rows in set (0.000 sec)

No 8 Query
(as Plain Text)

```
SELECT DISTINCT c.customer_name
FROM customer c
JOIN depositor d ON c.customer_id = d.customer_id
JOIN account a ON d.account_number = a.account_number
JOIN borrower b ON c.customer_id = b.customer_id
JOIN loan l ON b.loan_number = l.loan_number
WHERE a.branch_name = 'Perryridge'
AND l.branch_name = 'Perryridge';
```

No 8 SS (of Query & Output in Shell)	<pre> MariaDB [bank_23201336]> SELECT DISTINCT c.customer_name -> FROM customer c -> JOIN depositor d ON c.customer_id = d.customer_id -> JOIN account a ON d.account_number = a.account_number -> JOIN borrower b ON c.customer_id = b.customer_id -> JOIN loan l ON b.loan_number = l.loan_number -> WHERE a.branch_name = 'Perryridge' -> AND l.branch_name = 'Perryridge'; +-----+ customer_name +-----+ Hayes +-----+ 1 row in set (0.000 sec) </pre>
No 9 Query (as Plain Text)	<pre> SELECT c.customer_name, SUM(l.amount) AS total_loan FROM customer c JOIN borrower b ON c.customer_id = b.customer_id JOIN loan l ON b.loan_number = l.loan_number GROUP BY c.customer_name HAVING COUNT(l.loan_number) >= 2; </pre>

No 9 SS (of Query & Output in Shell)	<pre> MariaDB [bank_23201336]> SELECT c.customer_name, SUM(l.amount) AS total_loan -> FROM customer c -> JOIN borrower b ON c.customer_id = b.customer_id -> JOIN loan l ON b.loan_number = l.loan_number -> GROUP BY c.customer_name -> HAVING COUNT(l.loan_number) >= 2; +-----+-----+ customer_name total_loan +-----+-----+ Smith 2900 +-----+-----+ 1 row in set (0.000 sec) </pre>
No 10 Query (as Plain Text)	<pre> SELECT d.customer_id, SUM(a.balance) AS total_balance FROM depositor d JOIN account a ON d.account_number = a.account_number GROUP BY d.customer_id HAVING COUNT(d.account_number) > 1; </pre>
No 10 SS (of Query & Output in Shell)	<pre> MariaDB [bank_23201336]> SELECT d.customer_id, SUM(a.balance) AS total_balance -> FROM depositor d -> JOIN account a ON d.account_number = a.account_number -> GROUP BY d.customer_id -> HAVING COUNT(d.account_number) > 1; +-----+-----+ customer_id total_balance +-----+-----+ C-226 1400 +-----+-----+ 1 row in set (0.000 sec) </pre>

